UNIVERSITI TEKNOLOGI
MARA FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	MOCK 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack

   consists of: i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

QUESTION 1

a) Define database, DBMS and database system.

(6 marks)

b) State FOUR limitations of the file-based approach that a DBMS addresses.

(8 marks)

c) Nusantara Learning Hub still keeps its customer records in a spreadsheet program. Table 1 shows a sample of the data. Study Table 1 and answer the following.

Table 1 : Sample data for a spreadsheet program

	A	B	C	D	E
1	**Customer No**	Customer IC	First Name	Last Name	e-mail
2	100	2021877889	Abdul	Badri	a.abdul
3	101	2021755266	Anwi	BenBrahim	e.anwi
4	102	2021522447	Zainuddin	Hamid	zek90
5	103	2021369874	Kayrie	Jamlus	j.kayrie
6	104	2021123698	Huan	Ong	4d40ng
7	105	2021339693	Ishak	Sabarul	IS4u
8	106	2021745632	Rita	Hasrul	DOH

i) Identify TWO candidate keys and ONE primary key from Table 1. Give ONE reason for your chosen primary key.

(3 marks)

ii) Explain why the spreadsheet in Table 1 is not a substitute for a database.

(3 marks)

QUESTION 2

a) Nusantara University keeps its student and course records in the tables shown below. Table 1 shows the STUDENT table and Table 2 shows the COURSE table. Identify the primary key of each table and TWO candidate keys in the STUDENT table.

Table 1: STUDENT table

STUDENT_IC	STUDENT_ID	FIRST_NAME	LAST_NAME	EMAIL	COURSE_ID
80193939979	202587788	Ahmad	Ali	AA@gmail.com	40
40122566748	202575526	Haqiem	Rosly	haqiem@gmail.com	40
20164723670	202552244	Wany	Hazwan	wany@gmail.com	50
10108990074	202536987	Haliza	Husni	Ha@gmail.com	50
80116756736	202512369	Heechul	Kim	heechul@gmail.com	40
90137476323	202533969	James	Reece	james@gmail.com	20
50143670971	202574563	Canos	Sergi	Ola21@gmail.com	20

Table 2: COURSE table

COURSE_ID	COURSE_NAME
10	Networking
20	Software Developer
30	Web Designer
40	Database Developer
50	Data Analyst
60	IT Professional

(4 marks)

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b) Explain entity integrity and referential integrity. Give ONE example from Table 1 and Table 2.

(6 marks)

c) Define tuple, attribute, degree and cardinality.

(4 marks)

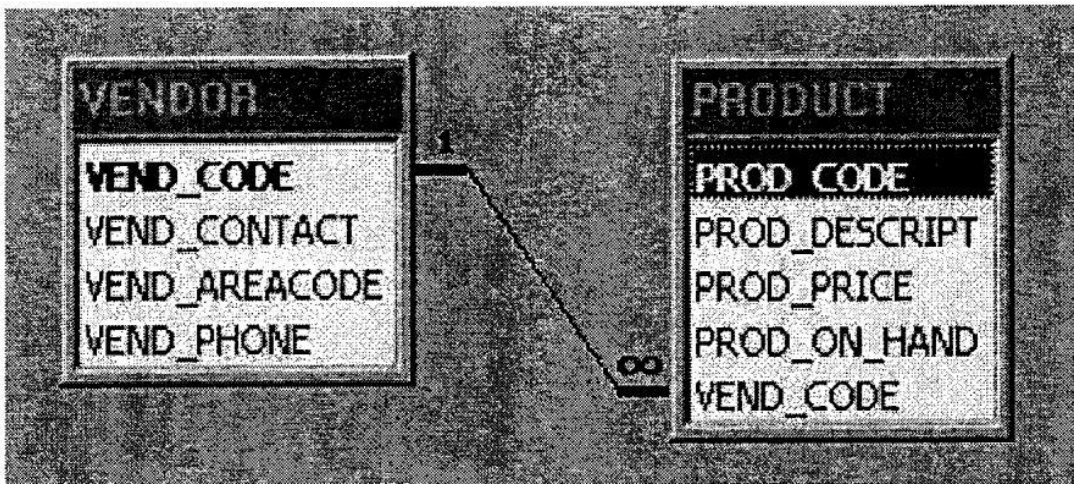
d) Explain TWO update anomalies that can occur in the STUDENT table if course details are stored together with student details.

(6 marks)

QUESTION 3

a) Figure 1 shows part of an ERD for a supplier and product database. Identify the entities, their attributes, the relationship between them and the cardinalities shown.

Figure 1 : Referential integrity in an Entity Relationship Diagram



(8 marks)

b) Explain the referential integrity constraint shown in Figure 1. Give ONE example of an operation that would violate it.

(6 marks)

c) Map the ERD in Figure 1 to a relational schema. Mark the primary key and foreign key.

(6 marks)

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QUESTION 4

a) The spreadsheet in Table 1 shows course and student records for a college. Study Table 1 and identify the candidate key and all functional dependencies.

Table 1 : Sample of college course information

Course Code	**ICT450**	**Course Name**	**Database Design and Development**	
Tutor ID	306612	Tutor Name	Professor Albus Dumbledore	
Student ID	Student Name	DOB	Gender	Last Attendance Date
2016123456	Luke Skywalker	11.10.96	F	31.01.18
2016534363	Kylo Ren	12.12.96	F	15.01.18
2016788924	Thor Loki	27.06.96	F	31.01.18
2016329874	Clint Barton	31.08.96	F	17.01.18
2016324793	General Hux	23.05.96	F	23.01.18
2016479625	Leia Organa	17.02.96	M	15.01.18
2016328796	Hermione Granger	15.04.96	M	31.01.18

(4 marks)

b) Explain 1NF, 2NF and 3NF.

(6 marks)

c) Identify the partial and transitive dependencies in Table 1.

(4 marks)

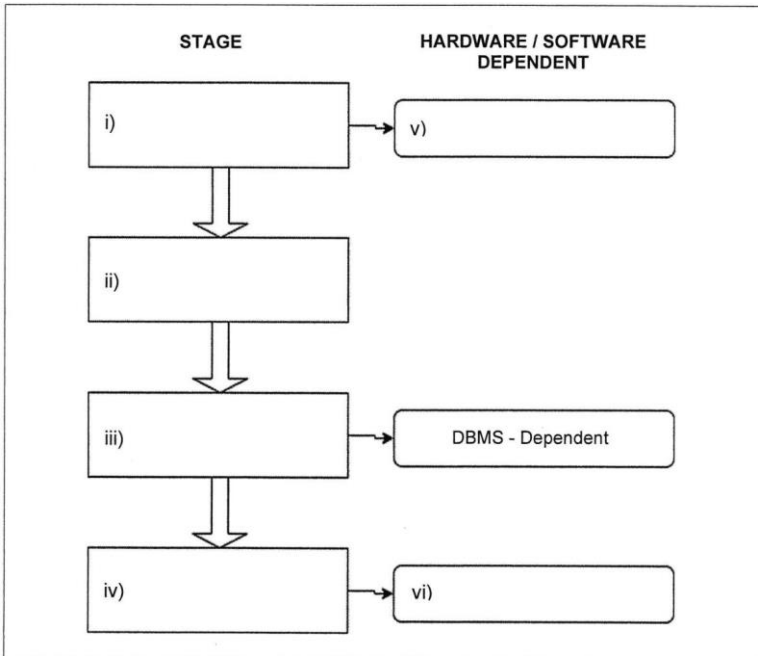
d) Decompose Table 1 into 3NF relations. Show the primary and foreign keys.

(6 marks)

QUESTION 5

a) Figure 1 shows the database design process. Explain FOUR activities carried out during the database design phase.

Figure 1 : Database Design Process



(8 marks)

b) Differentiate the SDLC and the DBLC.

(4 marks)

c) Compare top-down and bottom-up database design.

(4 marks)

d) Explain centralized and decentralized design.

(4 marks)

END OF QUESTION PAPER

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